ONE IMAGE, FIVE CAPTIONS

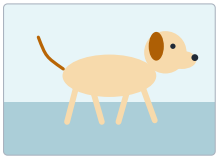


image x

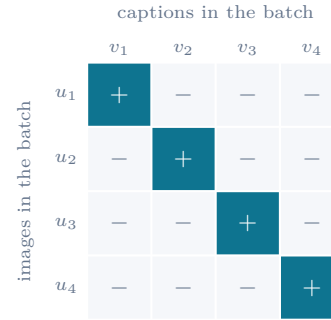
image encoder f_I → u

unit vector; each caption goes through the text encoder: $v_k = f_T(\text{caption } k)$

candidate caption k	cosine $u^\top v_k$	$p(k \mid x)$, $\tau = 0.01$
A a dog running on grass	0.31	0.84
B a car parked by the road	0.12	0.00
C a dog on grass, three people behind	0.28	0.04
D a golden retriever	0.29	0.11
E a bird in a forest	0.10	0.00

C mentions people who are not in the image, yet its cosine is almost as high as A. Dividing by τ turns the small cosine gaps into a sharp choice.

TRAINING: $B \times B$ SCORES



diagonal: true pairs, pushed up
off-diagonal: other pairs in the batch, pushed down

softmax along each row and each column